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A bioblitz provides valuable biodiversity data

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A bioblitz provides valuable biodiversity data

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Abstract

Bioblitzes are increasingly popular participatory science events aimed at documenting as many species as possible over a pre-defined period. They enhance community engagement and increase scientific literacy, while also contributing valuable biodiversity data. Bioblitzes can generate immense amounts of biodiversity data in a relatively short time span. But the value of such intense data are rarely quantified. In this study, we use the DeLuca Bioblitz as a case study to develop a biodiversity utility framework and a list of biodiversity metrics to evaluate the biodiversity contributions from bioblitzes. We calculate summary statistics from iNaturalist data collected during the bioblitz, including the total number of observations, number of observers, and number of species from the DeLuca Bioblitz. Then we compare these data to random polygons of the same area across the region, to evaluate the impacts of bioblitz efforts compared to participatory science generally. We found that bioblitzes tend to generate more biodiversity data when compared to other random polygons of the same size throughout the state of Florida, especially when considering the temporal comparison. Our results illustrate the potential of bioblitzes to generate valuable biodiversity data. The bioblitz in our case study also generated key records of new invasive species, new continental records, and records of imperiled species in Florida. Together, these data can help land managers and conservation professionals collect robust data, over a short duration, to create baseline species inventories and document species of concern for both management and conservation.

Keywords: community science; citizen science; adaptive management; bioblitz; iNaturalist

Introduction

Site-level species inventories, such as a list of possible species in a certain property or park, often serve as baseline data for natural resource managers (Maes et al. 2015; Katzer et al. 2025). This information is essential for natural resource managers to prepare management plans and monitor ecological changes over time, particularly as species shift their ranges (Weiskopf et al. 2020) and local extinction and colonization events become increasingly common (Cowie et al. 2022; Storch et al. 2022). However, gathering such data can be time-consuming and costly; therefore, inventories can be difficult to maintain (Katzer et al. 2025). Advances in technology have given rise to many long-term monitoring platforms, including participatory citizen science, sometimes referred to as community science or citizen science. Participatory citizen science is increasingly contributing to biodiversity research (Fauzia et al. 2023; Mason et al. 2025a), conservation (Hsu et al. 2025), and natural resource management (McKinley et al. 2017; Mason et al. 2025b). The bulk of all biodiversity data in the world now comes from platforms such as eBird and iNaturalist—upwards of 80% of all global biodiversity data since 2010 (Callaghan et al. 2020). Citizen science data are increasingly used to answer questions about species distributions (Zulian et al. 2021), to monitor biodiversity in a given geographic region (Tiralongo et al., 2020), and to detect biological invasions (Larson et al. 2020).

Bioblitzes are one mechanism in which participants are encouraged to contribute data through participatory science. A bioblitz has many definitions (Meeus et al. 2023), but in general, it involves a set time period where the goal is to document as many species as possible within a pre-defined geographic region (Meeus et al. 2023). This can range from a given park, or to large geographic regions. One particularly well-known example is the City Nature Challenge — a

four-day bioblitz organized annually by the California Academy of Sciences — where cities across the world participate in a friendly competition to get the most observations, species, and people using iNaturalist (Palma et al. 2024; Kishimoto & Kobori 2021; Sakurai et al. 2022). Bioblitzes are useful for engaging the public in conservation and creating a sense of personal involvement in environmental research (Roger & Klistorner 2016). They are also useful for generating large amounts of data in a given location (Meeus et al. 2021, 2023). The City Nature Challenge alone provides the most data of any week to the iNaturalist platform each year (Meeus et al. 2023; Palma et al. 2024). However, the majority of bioblitzes occur at localized scales, where potential participants are readily engaged (Young et al. 2021). Yet their scientific value is less often emphasized, particularly in terms of the biodiversity data they can generate for a specific site.

Bioblitzes can have a variety of outcomes, both for incentivizing people to continue using platforms like iNaturalist and also for their biodiversity data (Meeus et al. 2021, 2023). Bioblitzes build community, increase engagement in science, and serve as educational tools (Pollock et al. 2015; Postles and Bartlett 2015). They also contribute to species inventories, but less is known about other types of biodiversity data they help generate. Bioblitzes may be beneficial for biodiversity monitoring by detecting rare and cryptic species, new invasive species to a region, range shifts in imperiled species, or help filling gaps in a species' geographic distribution. These metrics extend beyond what can be examined with species lists, including invasive species management targets or endangered species range expansions. In this way, bioblitzes may offer value to land managers, conservation agencies, and academic institutions.

Therefore, it is critical to uncover the varied biodiversity utility that bioblitzes offer beyond simple species inventories for both land managers and conservation practitioners alike.

Here, we evaluate a bioblitz case study of a site (DeLuca Preserve, FL, USA) and empirical data analysis illustrating how this bioblitz has been useful in generating valuable biodiversity data. We illustrate the scientific value of bioblitzes as a mechanism for generating biodiversity data, including baseline species inventories, novel species records, and indicators of regional biodiversity utility (Figure 1). In particular, we highlight how concentrated, short-term efforts can uncover locally novel records and contribute to site-level species inventories in ways that complement broader-scale biodiversity monitoring. We provide a replicable framework that can be expanded outside of our study site to either a broader scale or a different bioblitz region, with the aim of helping natural resource managers leverage bioblitzes to advance biodiversity generation.

Methods

Study area

The DeLuca Preserve (27.670592, -80.952303) is situated in Osceola County in central Florida. The DeLuca Preserve is located near Yeehaw Junction, FL, a census-designated place in southern Osceola County about three hours south of the main University of Florida campus in Gainesville, FL, and about three hours north of the Miami metropolitan area in South Florida. The preserve was donated to the University of Florida by Elisabeth DeLuca in 2020 to place the land in permanent conservation (in collaboration with Ducks Unlimited) and to provide high-quality habitat for student research. DeLuca Preserve is ~11,000 hectares and consists of pine

flatwoods, Florida scrub, hardwood hammocks, forested and emergent wetlands, imperiled dry prairie, former citrus groves, and working cattle ranchland. The preserve hosts several state and federally listed species, such as the Florida grasshopper sparrow (*Ammodramus savannarum floridanus*), red-cockaded woodpecker (*Leuconotopicus borealis*), gopher tortoise (*Gopherus polyphemus*), and Florida panther (*Puma concolor couguar*). The area is now managed by the University of Florida Institute for Food and Agricultural Sciences (UF/IFAS) with varied management activities including active cattle ranching, prescribed burning in the pine flatwoods and pastures, and conserving waterfowl in the impounded wetlands on the property. The property has hunting leases.

The DeLuca Bioblitz

The DeLuca Bioblitz began as an initiative from the University of Florida Institute of Food and Agricultural Sciences (UF/IFAS) to promote public engagement with the newly established preserve while showcasing a conservation focal area of the Everglades Headwaters National Wildlife Refuge and Conservation Area. This annual event is designed as a free-ranging bioblitz, meaning participants are free to sample taxa and habitats of interest without any restrictions. However, we note that in the early years, there were some concerted efforts to obtain baseline biodiversity metrics with limited resources and people, which meant targeting specific portions or habitats of the property.

To date, the DeLuca Bioblitz has been running for four years and is expected to continue in future years. The dates of the previous bioblitzes were as follows: 26 February 2022, 24 February 2023, 1 March 2024, and 7 March 2025. Over the course of these four years,

participants have generated 13,161 observations from 1,252 species, with contributions from 203 unique individuals, hereafter referred to as observers. With its expanding datasets and consistent public involvement, the DeLuca Bioblitz is positioned to play a long-term role in documenting biodiversity trends and fostering awareness of the ecological significance of the DeLuca Preserve.

iNaturalist

iNaturalist is a citizen science platform where users can post and identify photographic or audio observations of any organism. It is accessible via the iNaturalist website or through their mobile application. As of October of 2025, iNaturalist has over 3 million observers and more than 2 million observations. After an observation has been uploaded with a species identification on iNaturalist, other users can agree or disagree with the identification. Once an observation reaches a two-thirds community consensus, it is considered ‘Research Grade’. Research Grade observations are made available for download on iNaturalist and are uploaded to the Global Biodiversity Information Facility (GBIF), where they are made available to scientists around the world. The DeLuca Bioblitz used iNaturalist as the central tool to collect and engage with participants of the bioblitz, where only observations from iNaturalist were included in the bioblitz results. We chose iNaturalist as our bioblitz platform because it is free, has many users, is user-friendly, and the resulting data are increasingly used in ecological research (Mason et al., 2025b). At check-in, participants were asked if they were familiar with iNaturalist, and those who were unfamiliar with the platform were given a brief tutorial on how to download and engage with the app.

Analysis of records

We obtained all iNaturalist data for the DeLuca Bioblitz efforts by using the iNaturalist data export tool on July 1, 2025. We aggregated the data from the four different bioblitz efforts, described above. For each species with a Research Grade observation(s), we also collected the total number of observations of that species in (a) Osceola County, (b) the state of Florida, and (c) the globe using the iNaturalist API (iNaturalist 2025). We report the relative frequency of these species found as part of the bioblitz efforts relative to the total number of observations of each species in Osceola County, as well as the state of Florida. Additionally, we used the iNaturalist API to obtain species origin information in relation to Florida (i.e., introduced, endemic) for all species recorded during DeLuca Bioblitzes. To assess the extent to which the bioblitz filled in locations in species distributions, for every species with fewer than 100 observations from the DeLuca Bioblitz, we found the next closest Research Grade observation outside of DeLuca Preserve. To do this, we used all Research Grade iNaturalist data in Florida through June 27th, 2025, downloaded using the iNaturalist data export tool. While using the Global Biodiversity Information Facility (GBIF) is the recommended approach for obtaining Research Grade data, we noticed that the taxonomy did not fully align between iNaturalist-exported data and GBIF. Therefore, we chose to use iNaturalist data export, so our methods aligned with our methods for obtaining the DeLuca Preserve data.

We also compared the cumulative spatial sampling effort from all of the DeLuca Bioblitzes to other randomly distributed locations across Florida. To do this, we used the create_square function in the sf package (Pebesma and Bivand 2023; Pebesma 2018) and randomly generated 100 different polygons (i.e., square boundaries) in the state of Florida that matched the size of

the DeLuca preserve (Figure S1). In each of these randomly generated polygons, we gathered the number of observers, number of observations, and number of species documented at Research Grade level using the iNaturalist API. Comparisons between the DeLuca Bioblitzes and the randomly generated polygons did not have the same temporal resolution. The DeLuca Bioblitzes cumulatively represented data collected on 4 days from 4 separate years, whereas the randomly generated polygons included all-time records (i.e., there was no temporal filter applied to the polygon-generated data).

Data availability

All data and code are available in a Github repository (https://github.com/coreytcallaghan/DeLuca_bioblitz_analysis) and will be archived in Zenodo.

Results

Overall contributions summary

At the DeLuca Preserve, we have hosted four bioblitzes, totaling 203 observers, 13,163 observations, and 892 research grade species. Throughout the four years, the number of unique observers has continued to grow, with 59 observers in the first year, 29 new observers in the second year, 28 new observers in the third year, and 87 new observers in the last year. Only two observers were present at every single bioblitz thus far. The most commonly reported species were pink sundew (*Drosera capillaris;* 210 observations), fourpetal St. Johnswort (*Hypericum tetrapetalum;* 207 observations), shiny blueberry (*Vaccinium mrysinites;* 159 observations), saw palmetto (*Serenoa repens;* 155 observations), and orange milkwort (*Senega lutea*; 149 observations), with 45% Research Grade species reported 1 or 2 times, and 31% were

only documented by a single observation. The spatial distribution of observations was heterogeneous across the preserve, with a pronounced concentration in areas of high accessibility (Figure 2). Species accumulation curves revealed a pronounced long-tailed distribution of species detection frequency and a non-asymptotic trajectory, indicating that species richness continues to increase with each sampling effort and that sampling completeness has not yet been achieved (Figure 3)

Using iNaturalist data to define species origin, we also examined the accumulation of Florida-endemic and introduced species across the four bioblitzes (Figure S2). Both groups exhibited upward trajectories consistent with the overall species accumulation curve, indicating that continued sampling will likely reveal additional taxa. To date, 69 introduced and 20 endemic species have been recorded; however, these counts are provisional and likely underestimate true totals due to incomplete sampling, ongoing data curation, and a lack of true coherence with iNaturalist information on species status.

Gap filling

When comparing with randomly-sampled polygons, we found that the efforts associated with the DeLuca Bioblitz led to markedly higher observations per km² (118.0 vs. 20.1 ± 44.7; $t_{99} = -21.9$, $p < 0.001$) and species per km² (7.98 vs. 4.28 ± 3.77; $t_{99} = -9.80$, $p < 0.001$) than the statewide average, while observers per km² did not differ significantly (1.82 vs. 2.40 ± 3.79; $t_{99} = 1.52$, $p = 0.13$) (Figure 4).

For species with less than 100 observations recorded during DeLuca Bioblitzes, we found that the average median linear nearest distance of the species recorded outside of the DeLuca preserve was 7.043 km and ranged from 0.0035 to 7,595 km. The nearest observations represent common species such as white beggarticks (*Bidens alba*), ballmoss (*Tillandsia recurvata*), and Pickerelweed (*Pontederia cordata*). The species that was furthest away was false smut of palm (*Graphiola phoenicis*), a fungal leaf disease that affects palm species, where the nearest observation was in Lisbon, Portugal. Though there are no other Research Grade iNaturalist observations in Florida, this species has been known to occur in this region (Elliott 2005). Upon speaking with an expert in this field, they agree with the genus-level identification of this species, *Graphiola*, but did not agree that species-level identification is possible based on photographic evidence, given that there are multiple species of *Graphiola* worldwide (Piepenbring et al. 2012). Nevertheless, this is still the first observation of *Graphiola* on iNaturalist in the United States.

An analysis of relative rarity

Comparison with county- and state-level biodiversity baselines revealed that the DeLuca Bioblitzes contributed disproportionately to records of regionally rare and infrequently documented species. A subset of species recorded at DeLuca (2 species: *Condylostylus flavipes* and *Aethes triassumenta*) were observed only within the preserve and had not been documented elsewhere in Florida (Figure 5). Bioblitz-derived observations accounted for a disproportionately high share of county-level occurrence records, particularly within Osceola County, where many species were represented primarily or exclusively by DeLuca observations (Figure S3). Shannon's diversity index further confirmed high diversity of observations at the preserve (H' =

5.81), approaching statewide diversity (H' = 6.82), indicating that the bioblitzes captured both common and rare taxa in a relatively short time frame compared to the entire time frame for the 100 random polygons.

Discussion

Bioblitzes are increasingly popular events (Figure S4), concomitant with a general increase in participatory citizen science in general (Feldman et al. 2021). While there are many benefits of such events for both engagement and natural resource management (Mason et al. 2025b), we provide a framework and an assortment of biodiversity metrics that can be used to evaluate the utility of bioblitzes for collecting relevant biodiversity data. We evaluate potentially valuable data both for local, place-based adaptive management of natural resources, where, in this instance, observers found at least 33 endemic species and 71 introduced species. We also illustrate how data from a bioblitz can contribute to broad, landscape-scale knowledge about biodiversity, where we found that bioblitzes contribute relatively the same or greater knowledge in a 96-hour time frame as random polygons throughout Florida during their entire iNaturalist history. Overall, for the first time, we show how a concentrated effort organized through a bioblitz can contribute essential data for natural resource management.

The value of a given biodiversity observation is context-dependent. Biodiversity observations may be locally common species, new county occurrences, threatened species, or the first detection of a potential invader. However, we note that there are often context-dependent qualitative findings, where a single record of a specific species can be very valuable, such as a

finding of a non-native species (Yeshwanth et al. 2025) or of a species of conservation concern (Wilson et al. 2020; Mesaglio et al. 2021). The DeLuca Bioblitz uncovered several notable findings, including a critically imperiled species, the Florida black dark bee (*Stelis ater*), along with multiple threatened (e.g., *Hemiargus ceraunus*) and endemic species (e.g., *Lupinus floridanus*), underscoring the conservation value of this property and the importance of continued biodiversity monitoring through the bioblitz. Since bioblitzes concentrate effort in space and time, they are especially powerful in this regard, acting as early-detection systems that complement ongoing monitoring programs. Further, when enough observations accumulate at a local site, they provide the foundation for mapping biodiversity patterns with increasing resolution (e.g., Figure 1). As analytical approaches for citizen science data continue to advance (Beninde et al. 2023; Stewart et al. 2023), such datasets could allow researchers not only to document presence but also to estimate relative abundance and detect temporal trends. In this way, repeated and well-organized bioblitzes could form the basis of long-term biodiversity monitoring at the site scale.

At a broader scale, we situated the DeLuca Bioblitz efforts in a statewide context by leveraging a random polygon approach to compare the biodiversity data generated from a short-temporal scale bioblitz to all collected temporal data documented on iNaturalist around the state of Florida. This comparison highlights the distinctive value of the concentrated effort of a bioblitz. Our random polygon analysis (Figure 4; Figure S1) provides a novel way to benchmark bioblitz contributions against broader iNaturalist activity. These comparisons illustrate the intensity of participation these events can generate. In terms of observations and species per unit area, the DeLuca Bioblitz efforts contributed 487% greater observations per km^2 and 86.5% greater

species per km² than the mean of random polygons, despite the dramatically shorter time frame. This illustrates that bioblitzes can produce unusually dense datasets, creating snapshots of biodiversity that provide a complementary and powerful perspective. Given that bioblitzes can generate robust species inventories in 24 hours that rival entire temporal scales in our random polygons, land managers should consider the benefits of these non-traditional sampling events to create baseline species inventories. Lastly, conservation practitioners interested in both rare and potentially invasive species can leverage bioblitz events to document large amounts of data rapidly, compared to more traditional monitoring methods that can be more costly and time-consuming.

Our evaluation focused on quantitative metrics that are broadly applicable across bioblitz events, but this represents only one dimension of their value. Future work could strengthen these insights by directly comparing bioblitz outcomes with professionally collected survey data, assessing how well volunteer-collected records align with standardized monitoring (Katzer et al. 2025). Additional approaches might include examining taxonomic biases (e.g., toward charismatic or easily photographed taxa or taxonomic uncertainty), quantifying participant learning or engagement outcomes (Echeverria et al. 2021; Niemiller et al. 2021), or integrating bioblitz data with other monitoring tools such as eDNA or remote sensing (Granqvist et al. 2025; Cavender-Bares et al. 2022). Importantly, the results we present here emphasize short-term outcomes—the concentrated biodiversity snapshots generated over the course of a single event. The longer-term value, particularly the potential for repeated bioblitzes to build cumulative datasets that enable site-level monitoring and adaptive management, remains to be fully realized. As such, bioblitzes represent not only a means to engage the public but also an underutilized mechanism for

generating robust, repeatable biodiversity data with clear relevance for conservation and management.

Conclusions

We provide an overview of how a bioblitz can provide valuable biodiversity data. While we introduce quantitative metrics and provide a conceptual framework (Figure 1) for how data from a bioblitz can contribute to both local place-based natural resource management as well as landscape-scale biodiversity research. However, we stress that hosting a bioblitz is time-consuming and requires logistical planning (Tiago et al. 2024). Nevertheless, based on our results and collective experience, we suggest that natural resource managers should consider the use of bioblitzes and how they can contribute to adaptive management goals.

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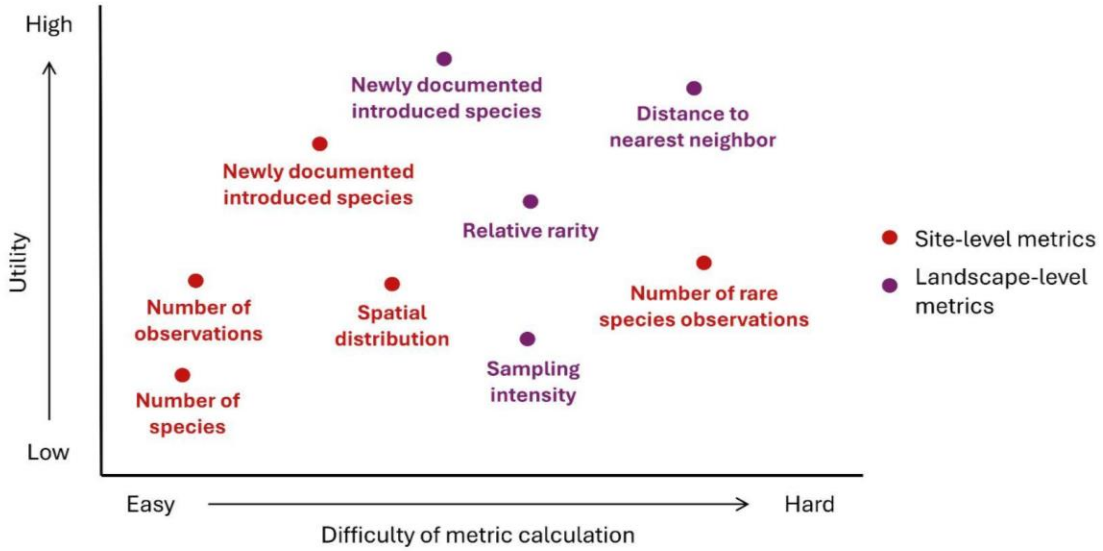
Figures

(A) Conceptual diagram



(B) Site-level metric definitions

Metric	Definition
Number of observations	Count of Research Grade observations in the site.
Number of species	Count of uniquely documented species in the site.
Spatial distribution	Density heat map of observations in the site.
Newly documented introduced species	List of non-native species that are documented at the site for the first time.
Number of rare species observations	List of species that are rarely documented in the site.

(C) Landscape-level metric definitions

Metric	Definition
Sampling intensity	Comparison of site-level observation and species counts to 100 random polygons in the region.
Distance to nearest neighbor	Distance of each site-level observation with less than 100 observation to nearest observation outside of site.
Newly documented introduced species	List of non-native species newly documented in the region.
Relative rarity	Comparison of site-level observation count to the observation count in the region.

Figure 1. (A) Conceptual diagram showing the utility and difficulty of calculating the nine biodiversity metrics explored in this paper. Metrics are grouped as either site-level (defined relative to the bioblitz site) or landscape-level (defined as the contribution of the bioblitz site to the broader region). (B) Definitions of site-level metrics based on Research Grade iNaturalist observations within the bioblitz site. (C) Definitions of landscape-level metrics, which incorporate Research Grade iNaturalist observations from outside the bioblitz site to represent the broader region.

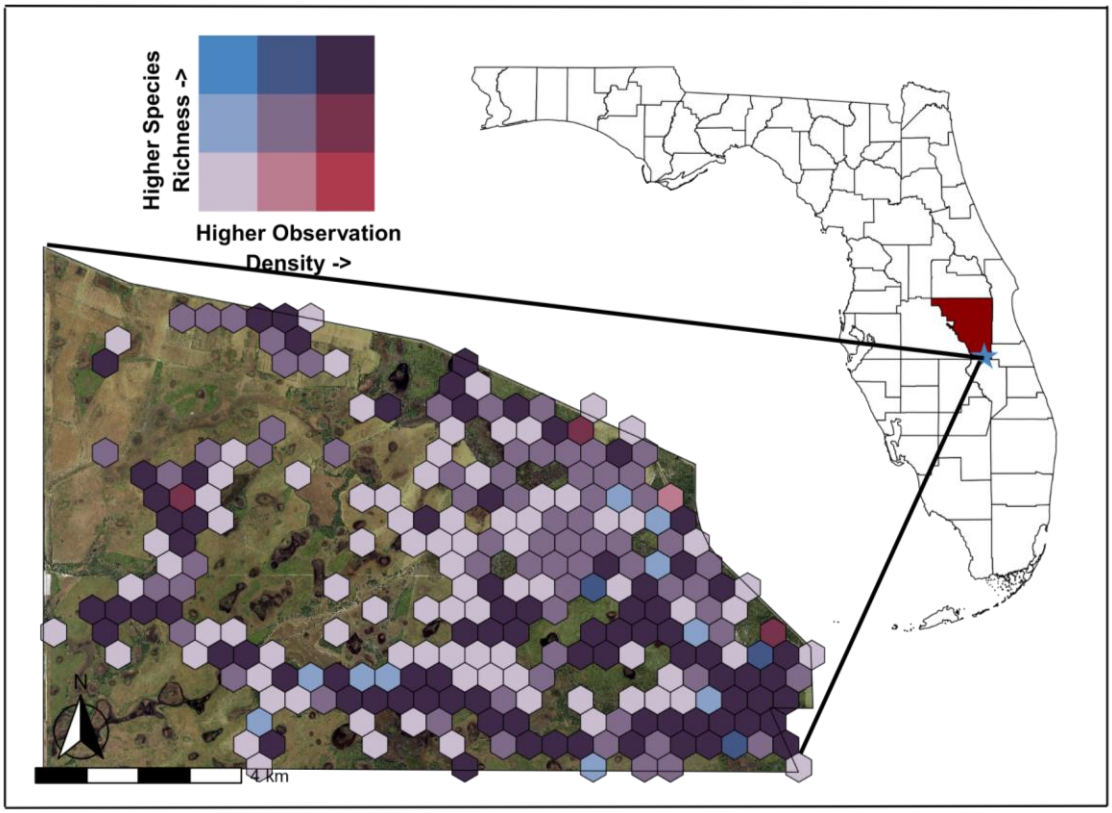


Figure 2. A map of DeLuca Preserve, showing its location in Florida and its relative remoteness. Osceola County, where the preserve is located, is highlighted in red. Additionally displayed is a density heat map of where the observations within the preserve have been derived from the DeLuca Bioblitzes, and the concentration of observations and species richness. Bivariate colors reflect quantile-based classification (dim = 3) of observation density and species richness.

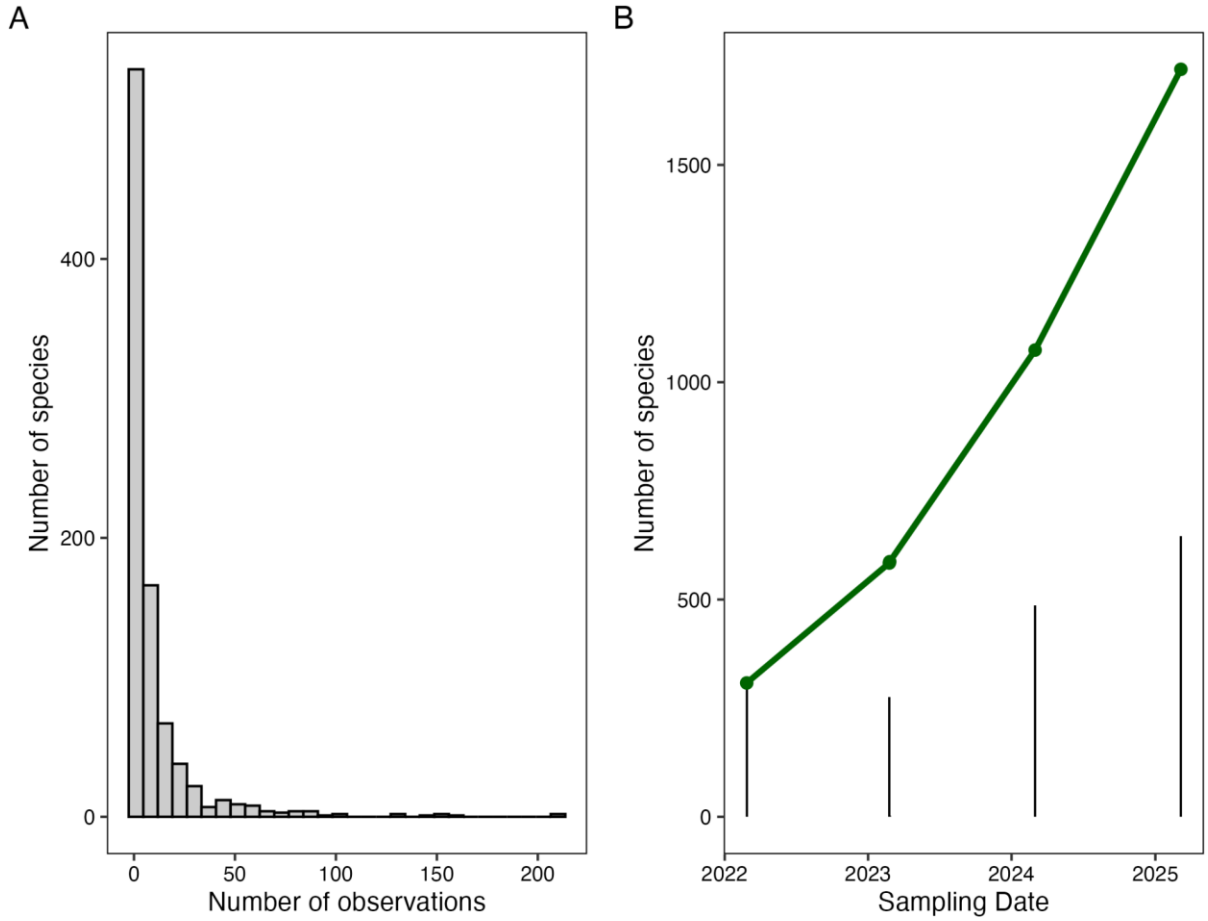


Figure 3. (A) Frequency distribution of species detection rates across the DeLuca Bioblitz dataset, showing the number of species in each observation group. (B) Species accumulation across sequential bioblitz events (2022–2025), indicating continued discovery of novel taxa and incomplete sampling saturation. The vertical lines represent the number of new Research Grade species found that year.

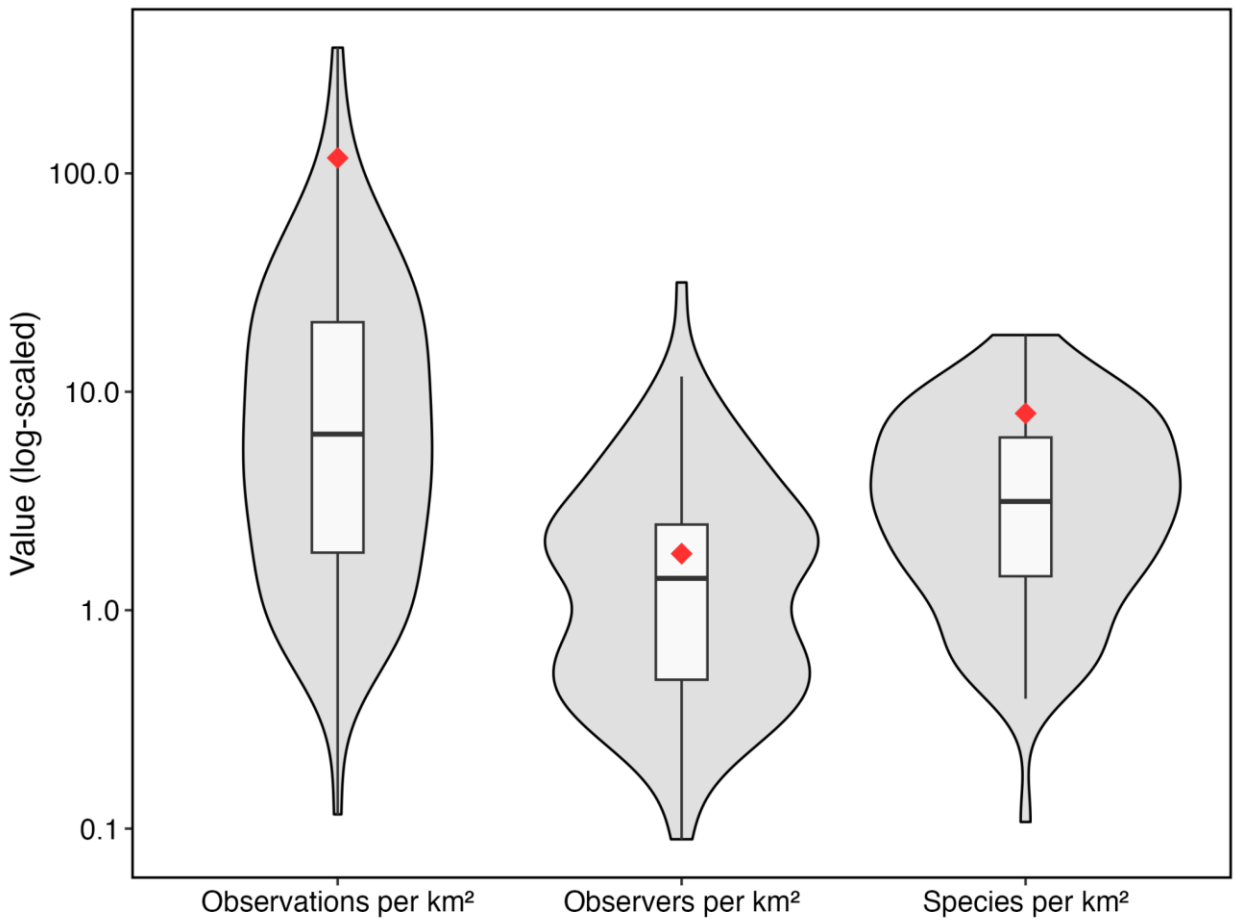


Figure 4. Comparison of biodiversity metrics at the DeLuca Preserve relative to 100 randomly selected polygons of equivalent area across Florida. Violin and boxplots represent the statewide distribution of iNaturalist data in random polygons for three metrics normalized by area (observers, observations, and species per km²), while the diamond markers indicate values from the DeLuca Bioblitz dataset. DeLuca exhibited markedly higher observation density (118.0 vs. 20.1 ± 44.7; $t_{99} = -21.9$, $p < 0.001$) and species density (7.98 vs. 4.28 ± 3.77; $t_{99} = -9.80$, $p < 0.001$) than the statewide average, whereas observer density did not differ significantly (1.82 vs. 2.40 ± 3.79; $t_{99} = 1.52$, $p = 0.13$).

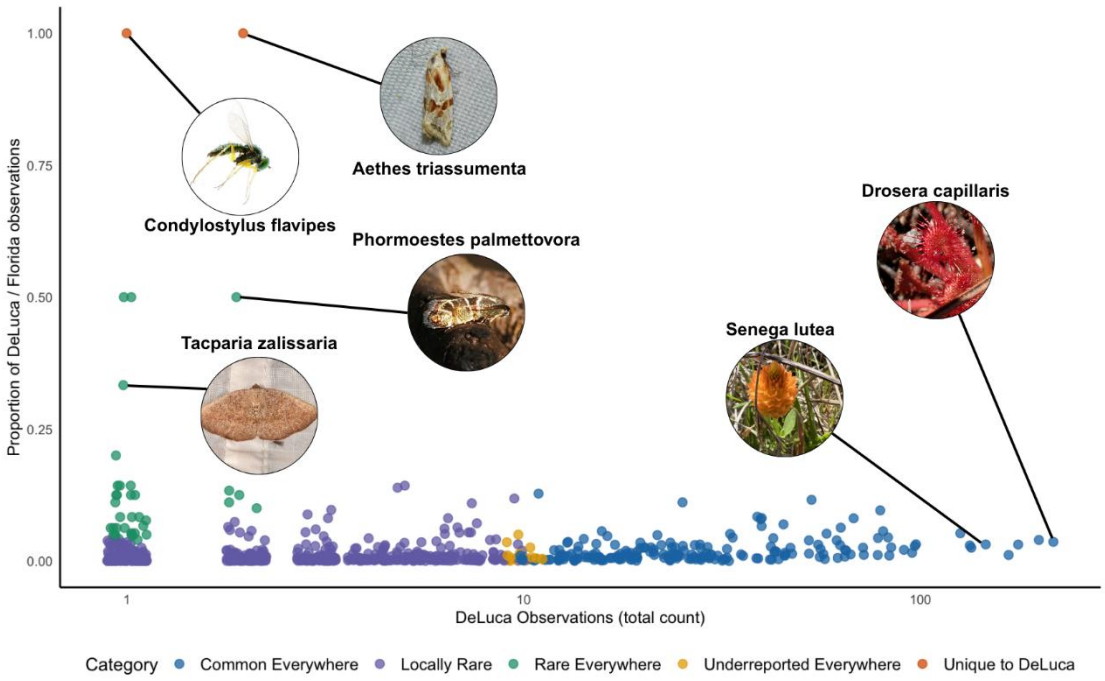


Figure 5. Each point represents a species observed at DeLuca, plotted by its total observation count at DeLuca (log-scale) versus its proportional representation relative to Florida-wide observations. Colors indicate rarity categories: blue represents common everywhere (> 10 observations in DeLuca; > 25 observations in Florida), yellow represents underreported everywhere (= 10 observations in DeLuca; <= 25 observations in Florida), purple represents locally rare (< 10 observations in DeLuca; > 25 observations in Florida), green represents rare everywhere (< 10 observations in DeLuca; < 25 observations in Florida); and red represents unique to DeLuca. Image attributions: Odin Toness (*Condylostylus flavipes*), Laura Gaudette (*Tacparia zalissaria*), Jeff Niznik (*Aethes triassumenta*), Liz Childress (*Phormoestes palmettovora)*, Aurora Alcaide (*Senega lutea*), and Donald Fraser (*Drosera capillaris*).

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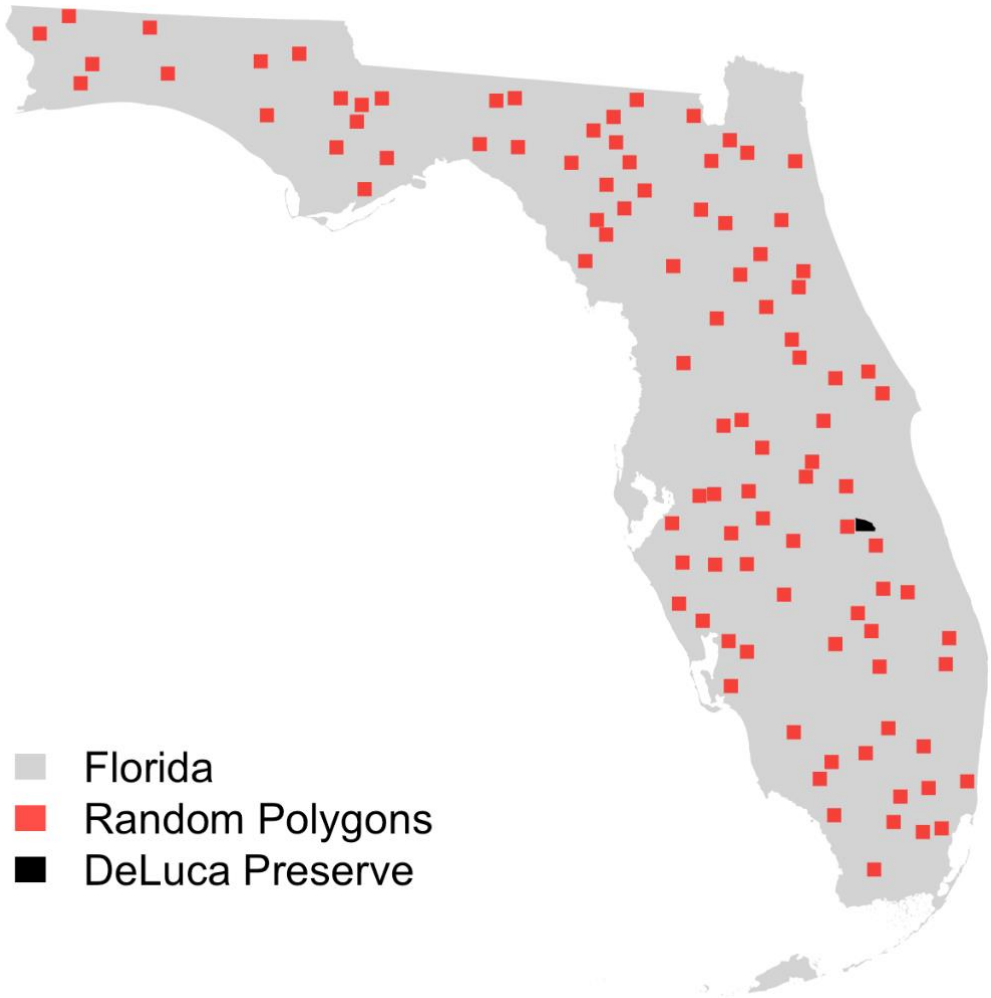

Figure S1. To determine the cumulative effort and illustrate the potential people-hours compared to other locations in Florida, we generated 100 random square boundaries with the same area as the DeLuca preserve across Florida.

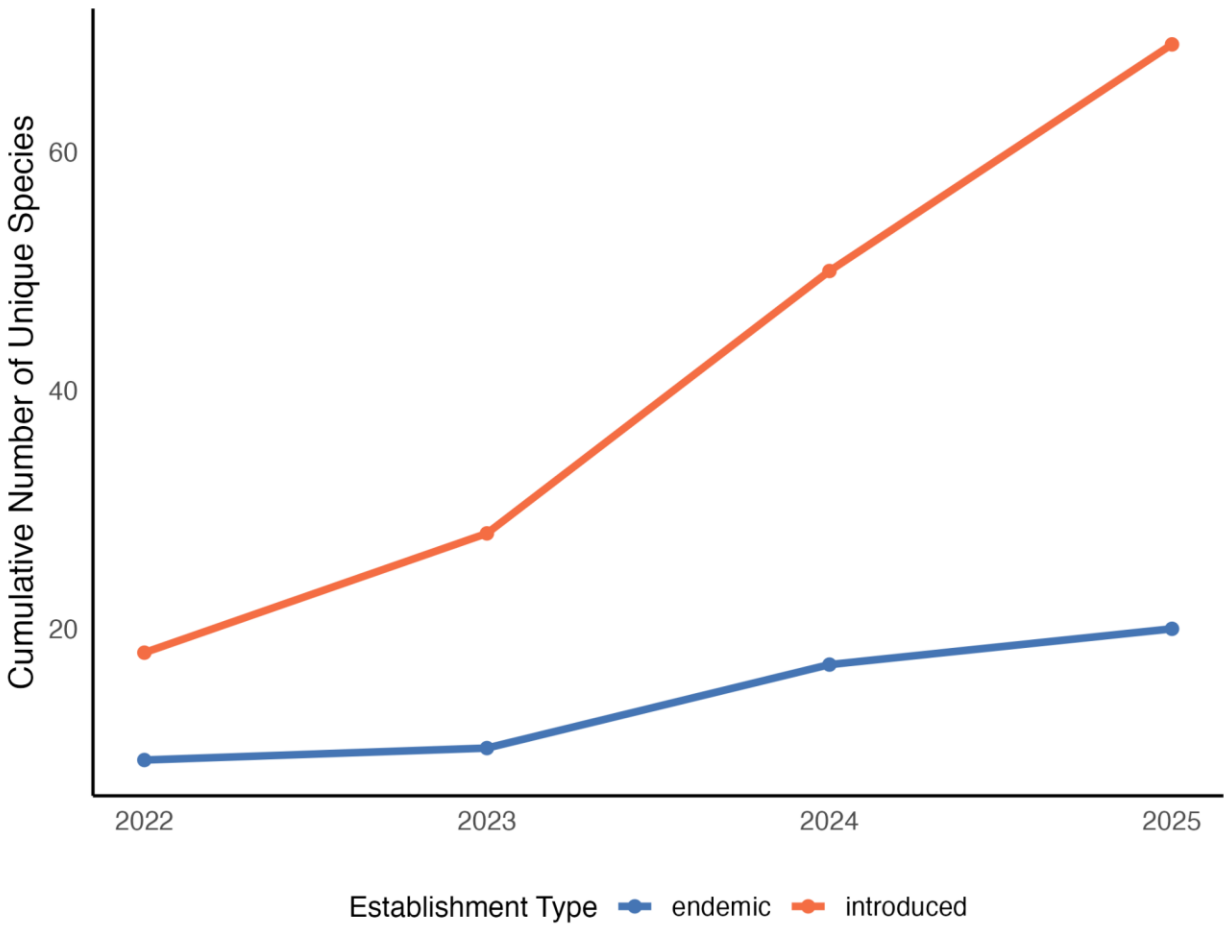

Figure S2. Cumulative number of endemic and introduced species recorded during bioblitz events at DeLuca Preserve, based on Research Grade iNaturalist observations. The orange line represents the number of introduced species over time. The blue line represents the number of endemic species to Florida over time. Both represent an increase in findings of establishment type per sampling effort.

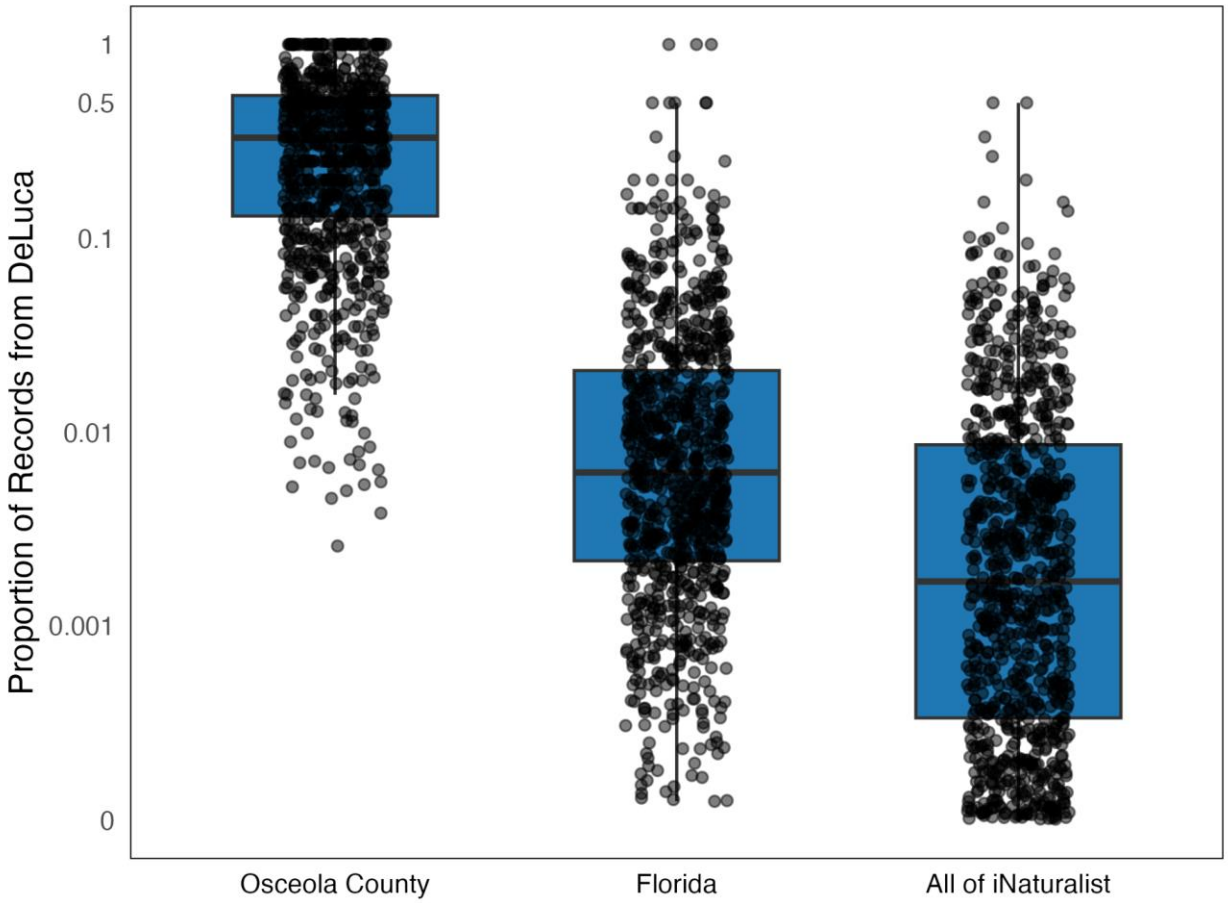

Figure S3. Proportion of all recorded observations per species that originated from the DeLuca Bioblitzes across different spatial scales. Each point represents a single species, with boxplots denoting the interquartile range. The three panels correspond to (from left to right): observations within Osceola County, all observations across Florida, and all observations on iNaturalist. All observations are Research Grade.

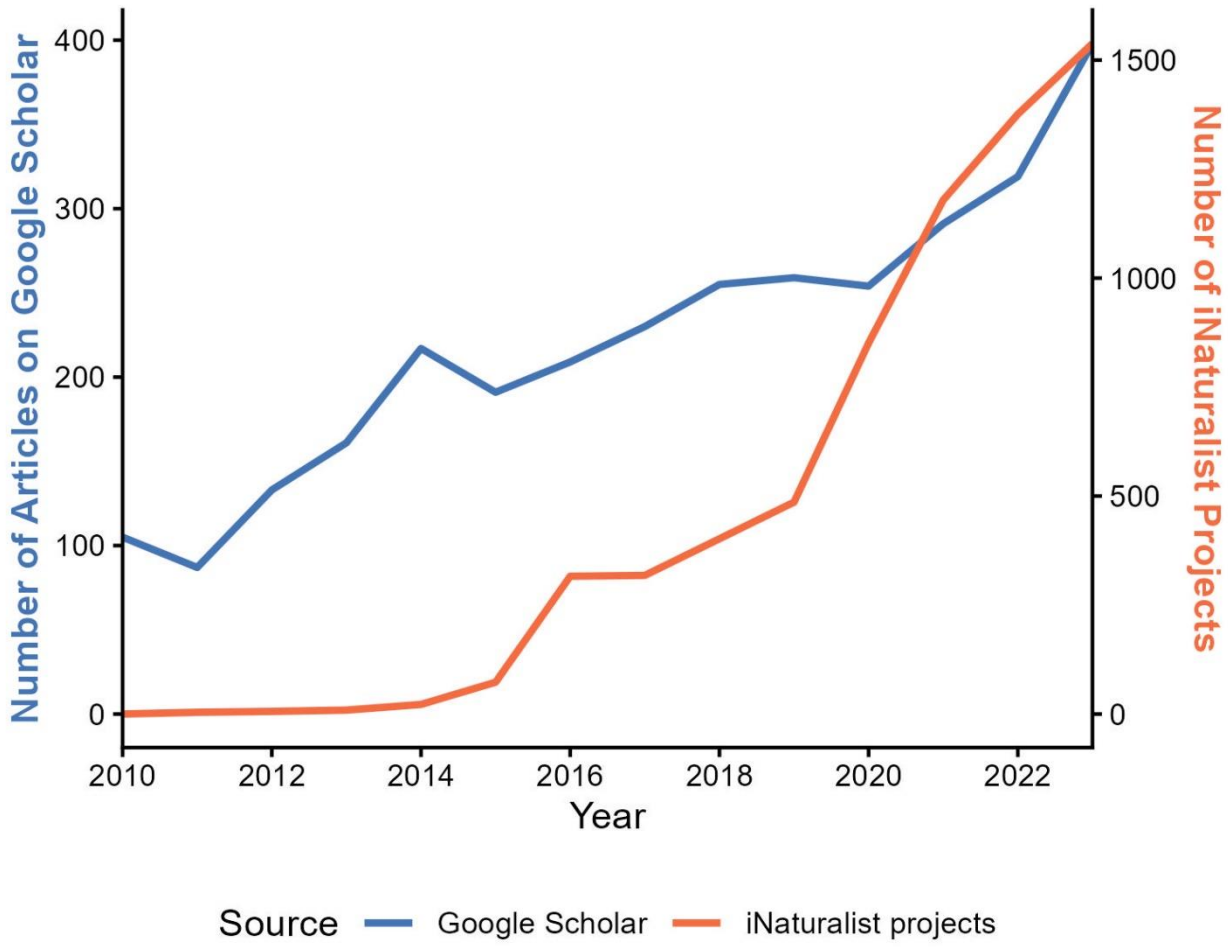

Figure S4. The blue line represents the number of articles returned from Google Scholar searches for the term "Bioblitz" by year. The orange line represents the number of iNaturalist projects that contain the term "Bioblitz" by year. Both cases illustrate the increasing popularity of bioblitzes over time.